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Electroencephalogram (EEG) findings in patients diagnosed with COVID-19

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Abstract

Background COVID-19 is known to affect multiple organ systems, including the brain, leading to neurological complications such as seizures, encephalopathy, and cognitive dysfunction. However, its impact on brain electrical activity remains unclear. Electroencephalography (EEG) is a valuable tool for detecting brain dysfunction, particularly in critically ill patients.

Methods A case-control study was conducted involving 95 patients presenting with respiratory symptoms suggestive of recent COVID-19 infection. Participants were classified into two groups: the COVID-19 group (study group) ($n=49$), consisting of patients with a confirmed diagnosis of COVID-19 based on RT-PCR testing and chest CT findings, and the non-COVID-19 control group ($n=46$), comprising patients with similar clinical symptoms but negative COVID-19 test results. All patients underwent comprehensive neurological examinations and continuous electroencephalogram (EEG) monitoring for six hours at two time points: within 24 h of hospital admission and following clinical improvement prior to discharge.

Results EEG abnormalities were observed in 51.0% of patients in the COVID-19 group, significantly higher than the 28.3% observed in the non-COVID-19 group ($p < 0.05$). The most frequent EEG findings included generalized slowing, focal slowing, and rhythmic delta activity. Patients exhibiting EEG abnormalities experienced significantly longer hospital stays compared to those with normal EEG results (average 20.4 vs. 10.5 days, $p < 0.05$). However, no significant association was found between EEG abnormalities and the severity of pulmonary involvement ($p > 0.05$).

Conclusion COVID-19 is associated with EEG abnormalities, particularly diffuse slowing and rhythmic delta activity, which may indicate brain dysfunction. EEG could be a useful tool for monitoring neurological involvement in COVID-19, especially in patients with persistent symptoms.

Keywords COVID-19, EEG, brain function, EEG abnormalities, neurological complications

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Introduction

Coronavirus disease 2019 (COVID-19) is primarily known for affecting the respiratory system, but growing research indicates that it can also impact the nervous system, leading to a variety of neurological complications. Many COVID-19 patients have reported symptoms such as headaches, dizziness, cognitive dysfunction, and loss of smell or taste. In more severe cases, seizures, encephalopathy, and even strokes have been observed, particularly in hospitalized patients [1].

One of the most effective ways to study brain activity and detect neurological abnormalities is electroencephalography (EEG). EEG is a non-invasive technique that measures the brain's electrical activity, helping to identify conditions such as epileptic seizures, encephalopathy, and cerebral dysfunction. While EEG has been widely used to assess brain disorders, its role in understanding COVID-19-related neurological effects is still emerging [2].

Recent studies suggest that many COVID-19 patients show EEG abnormalities, including generalized slowing, rhythmic delta activity, and epileptiform discharges [3]. However, there is still limited data on whether these EEG changes are directly caused by the virus, secondary to hypoxia, inflammation, or metabolic disturbances, or simply a result of severe illness and hospitalization.

This study aims to investigate whether EEG abnormalities are more common in COVID-19 patients compared to those with similar respiratory symptoms but without COVID-19. Additionally, it seeks to determine if specific EEG patterns are linked to disease severity. We hypothesize that patients with more severe pulmonary involvement due to COVID-19 are more likely to exhibit EEG abnormalities, reflecting a potential association between respiratory disease severity and cerebral dysfunction,

which could help in identifying patients at higher risk for neurological complications. Understanding these relationships could improve early diagnosis, monitoring, and management of neurological symptoms in COVID-19 patients.

Subjects and methods

Study design and participants

This case-control study was conducted at Mataria Teaching Hospital between January 2023 and April 2024. A total of 95 patients presenting with chest symptoms suggestive of recent COVID-19 infection were enrolled. Common presenting symptoms included fever, dyspnea, cough, headache, malaise, anosmia, and generalized body aches. Participants were divided into two groups based on diagnostic findings. The COVID-19 group ($n=49$) included patients with confirmed COVID-19-related respiratory illness, diagnosed through two consecutive positive reverse transcription polymerase chain reaction (RT-PCR) tests and chest computed tomography (CT) scans reviewed by a pulmonologist. The non-COVID-19 control group ($n=46$) comprised patients with similar respiratory symptoms who were confirmed to have non-COVID-19-related respiratory illness, based on two negative RT-PCR tests and chest CT findings inconsistent with COVID-19 (Figs. 1, 2, 3, 4, 5 and 6).

The study was conducted in accordance with the Declaration of Helsinki. It was approved by the Ethics committee of the Faculty of Medicine, Ain Shams University and informed consent was obtained from every patient and control.

Inclusion and exclusion criteria

Eligible participants were adults aged 18 to 80 years, of both sexes, who were fully conscious, oriented, and

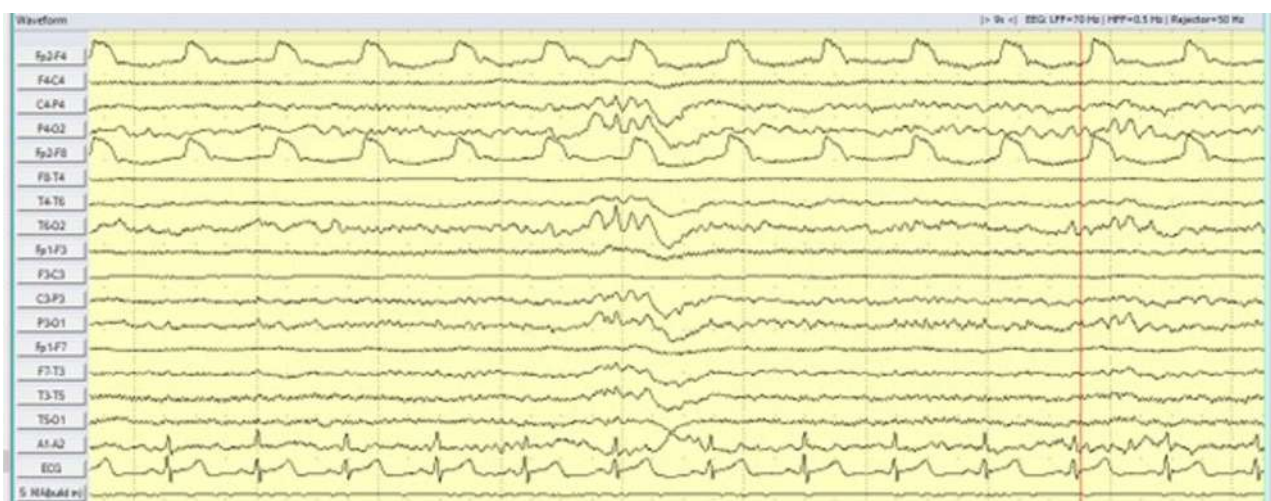


Fig. 1 Bilateral focal slowing EEG changes. A case of female patient 50 years old with history of diabetes mellitus and hypertension with confirmed recent Covid-19 infection, her CT chest showed CORADS scale=5, her first EEG showing Focal Slowing (Bilateral Temporo-parietal slowing theta range)

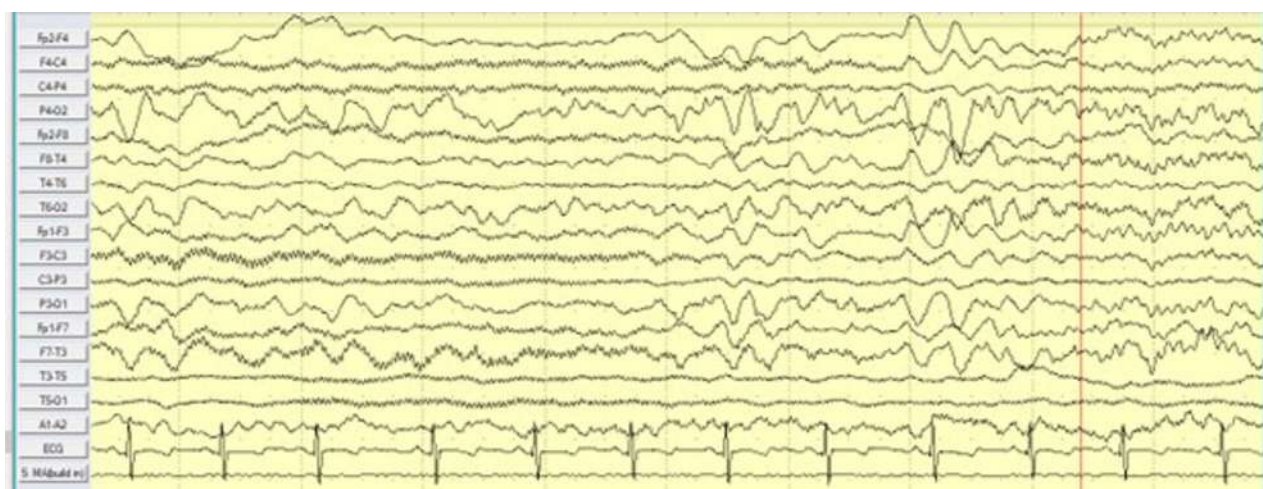


Fig. 2 Left side focal slowing EEG changes. Female patient 79 years old with history of Hypertension and confirmed recent Covid-19 infection, her CT chest showed CORADS scale = 4, her first EEG showing Left side Focal Slowing (Delta range)

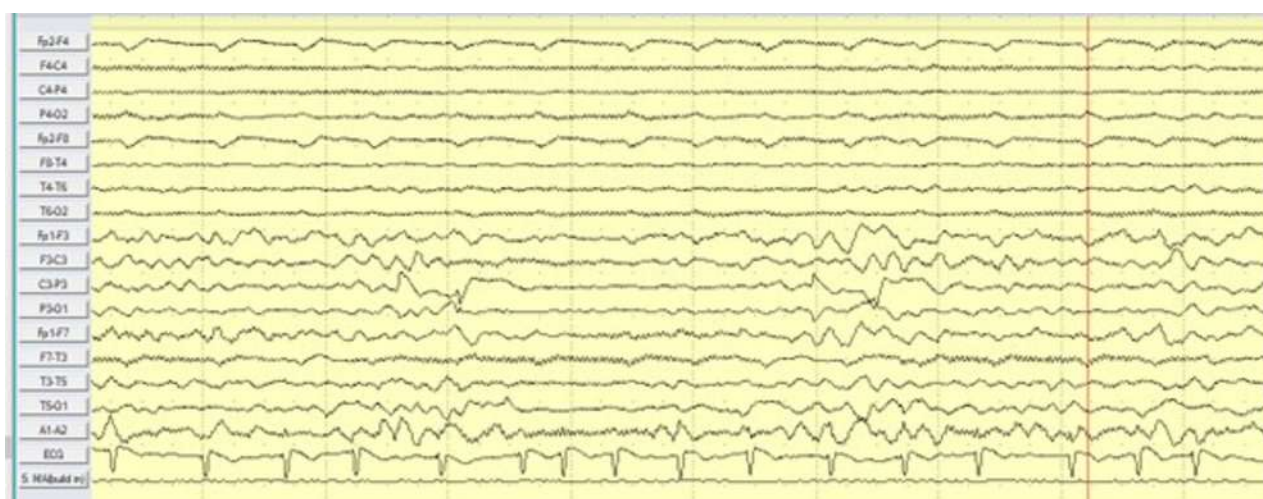


Fig. 3 Generalized slowing EEG changes. Female patient 38 years old with confirmed recent Covid-19 infection, her CT chest showed CORADS scale = 5, her second EEG showing generalized slowing

cooperative at the time of enrollment. Inclusion required recent hospital admission due to respiratory symptoms highly suggestive of recent COVID-19 infection, without any evidence of altered consciousness or overt encephalopathy.

Exclusion criteria included a prior diagnosis of epilepsy or seizure disorders, known brain malignancies, and severe metabolic conditions that may affect conscious level such as diabetic ketoacidosis, hepatic failure, or moderate to severe renal impairment (serum creatinine > 1.7 mg/dL). Patients with hypertensive encephalopathy, severe heart failure, major psychiatric disorders, primary headache syndromes, or any other condition likely to interfere with accurate EEG interpretation were also excluded from the study.

Clinical and neurological assessment

All patients underwent comprehensive clinical and neurological evaluations. This included a detailed medical history and general physical examination, followed by a full neurological assessment with particular attention to mental status, behavioral changes, and the presence or history of headaches. A thorough chest examination was conducted by a pulmonologist to evaluate the extent of respiratory involvement. Laboratory investigations were performed to exclude alternative causes of critical illness and included complete blood count (CBC), renal and liver function tests, HbA1c, and serum electrolyte levels. Radiological assessments, including chest computed tomography (CT) and brain magnetic resonance imaging (MRI), were also obtained as part of the diagnostic work-up.

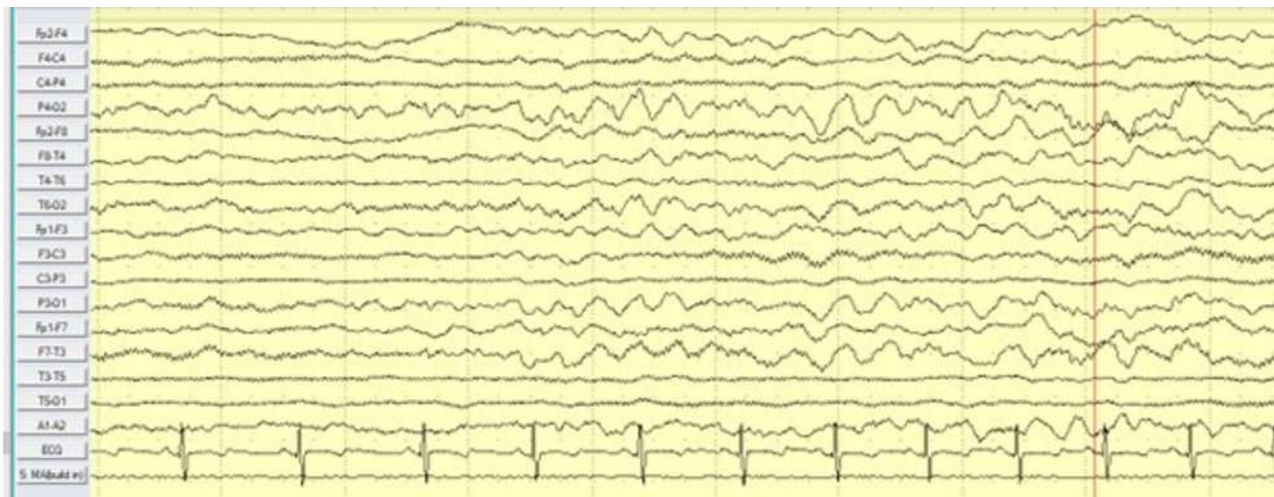


Fig. 4 Generalized rhythmic delta activity. Female patient 48 years old with history of Hypertension and confirmed recent Covid-19 infection, her CT chest showed CORADS scale=4, her second EEG showing Generalized Rhythmic Delta Activity



Fig. 5 Generalized epileptiform activity-1. Male patient 39 years old with medical history of cardiac disease and confirmed recent Covid-19 infection, his CT chest showed CORADS scale=5, his first EEG showing generalized paroxysmal epileptiform activity

COVID-19 severity classification using CO-RADS scale

Chest CT findings were evaluated and classified using the COVID-19 Reporting and Data System (CO-RADS) [4], a standardized scale for assessing the likelihood of COVID-19 pneumonia based on radiological patterns. The CO-RADS scale categorizes findings as follows: CO-RADS 1–2, indicating a low level of suspicion for COVID-19 pneumonia; CO-RADS 3, reflecting indeterminate findings; and CO-RADS 4–5, suggesting a high level of suspicion based on typical imaging features. For the purposes of this study, patients with CO-RADS scores of 3–5 were considered to have moderate to severe COVID-19 infection, whereas those with scores of 1–2 were classified as having mild or non-COVID-19-related respiratory illness.

EEG recording and analysis

Electroencephalographic (EEG) monitoring was performed for all patients at two predefined time points: the first EEG was conducted within the first 24 h of hospital admission, and the second EEG was performed following clinical improvement and prior to discharge. EEG acquisition was carried out using a standard 21-electrode setup in accordance with the International Federation of Clinical Neurophysiology (IFCN) guidelines and the international 10–20 electrode placement system [5].

All participants underwent continuous EEG monitoring for six hours using the Encephalan-131-03 EEG system (Medicom MTD, Taganrog, Russia). Recordings were conducted in a controlled EEG laboratory environment with regulated lighting and sound conditions to reduce external artifacts. Electrode impedance was maintained below 5 k Ω . EEG signals were recorded using

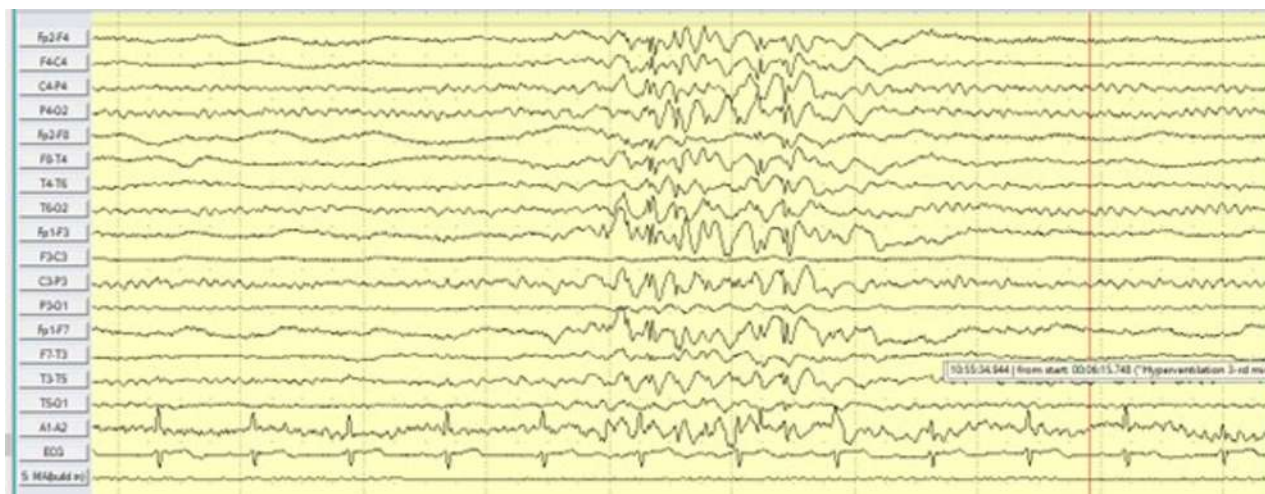


Fig. 6 Generalized epileptiform activity-2. Female patient 44 years old with medical history of hypertension and confirmed recent Covid-19 infection, her CT chest showed CORADS scale = 5, her first EEG showing Generalized paroxysmal epileptiform activity

Encephalan-131-03 EEGViewer version 3, with a sampling rate ranging from 250 to 1000 Hz and appropriate bandpass range (0.5–70 Hz).

Each EEG recording was reviewed separately, and critical epochs were analyzed independently by five experienced neurophysiologists who were blinded to clinical and group assignment data to ensure inter-rater reliability. Artifacts were manually identified and excluded from analysis. Any discrepancies in interpretation were resolved through consensus among the reviewing neurophysiologists.

Each EEG session consisted of six hours of continuous monitoring. Recordings were conducted using both bipolar and referential montages to enhance spatial resolution and diagnostic accuracy. Artifact removal was performed manually by two experienced epileptologists, who independently reviewed the recordings to identify and eliminate non-physiological signals. This approach ensured high-quality data and minimized the risk of misinterpretation due to external or physiological artifacts.

For each six-hour EEG session, critical segments were selected based on periods of wakefulness and minimal artifacts, ensuring representative sampling of the patient's cerebral activity. Transient abnormalities (e.g., brief focal slowing or isolated epileptiform discharges lasting < 10 s) were recorded but distinguished from persistent abnormalities, which were defined as recurrent or continuous EEG changes present across multiple epochs or maintained for > 50% of the recording duration. Persistent abnormalities were prioritized in the analysis as indicators of sustained cerebral dysfunction.

EEG findings classification

EEG findings were categorized into four primary types: background abnormalities, including generalized or focal

slowing; rhythmic delta activity, either generalized or lateralized; periodic discharges, classified as generalized or focal; and epileptiform abnormalities, encompassing focal or generalized epileptiform discharges. Each EEG recording was independently reviewed by five board-certified epileptologists who were blinded to the patients' COVID-19 status, ensuring objective interpretation and minimizing diagnostic bias.

Statistical analysis

Data analysis was performed using IBM SPSS Statistics, Version 24. Descriptive statistics were expressed as mean $\pm$ standard deviation (SD) for normally distributed continuous variables, and as median with interquartile range (IQR) for non-normally distributed variables. Categorical variables were summarized using frequencies and percentages.

Comparative analyses were conducted using the chi-square test for categorical data, independent t-tests for normally distributed continuous variables, and the Mann–Whitney U test for non-parametric continuous variables. Spearman's correlation coefficient was employed to assess the relationship between EEG abnormalities and the severity of respiratory involvement. A p -value < 0.05 was considered indicative of statistical significance.

Results

Demographic and clinical characteristics

The study included 49 COVID-19 patients and 46 non-COVID-19 patients. Regarding demographic data no statistically significant differences were found between the two groups. in terms of age ($p = 0.061$) and sex distribution ($p = 0.262$), Table 1.

Table 1 Comparison of Demographic Data Between COVID-19 and Non-COVID-19 Groups

Variable	Group 1: COVID-19 (n=49)		Group 2: Non-COVID-19 (n=46)		P-value
	No.	%	No.	%	
Sex					0.262
Male	21	42.9	25	54.3	
Female	28	57.1	21	45.7	
Age					0.061
Range	39–71		25–70		
Mean	58.7		55.0		
SD	7.5		9.4		
Comorbidities					
Hypertension	12	24.5	10	21.7	0.751
Diabetes mellitus	5	10.2	5	10.9	1.000
Cardiac disease	5	10.2	7	15.2	0.525
Renal disease	6	12.2	8	17.4	0.474
Hepatic disease	10	20.4	11	23.9	0.682

SD: standard deviation, COVID-19: Coronavirus Disease 2019

The two groups were well-matched regarding age and gender, ensuring that observed differences in EEG findings are not due to demographic variability.

EEG abnormalities in COVID-19 and non-COVID-19 patients First and second EEG findings

EEG abnormalities were significantly more frequent in COVID-19 patients compared to non-COVID-19 patients in both the first and second EEG assessments. The first EEG demonstrated a significantly higher incidence of abnormalities among COVID-19 patients compared to non-COVID-19 controls ($p=0.02$), with a notably greater frequency of generalized background slowing ($p=0.032$) and focal slowing ($p=0.045$). Persistent EEG abnormalities were observed in 28.6% of patients in the COVID-19 group, significantly higher than the 13.0% observed in the non-COVID-19 group ($p=0.05$). Although the overall prevalence of abnormalities decreased in the second EEG recordings, a substantial proportion of COVID-19 patients continued to exhibit persistent EEG changes, suggesting the possibility of ongoing or long-term neurological involvement, Table 2.

Table 2 EEG Abnormalities in First and Second EEG

EEG finding	1st EEG – COVID-19 group (n=49)	1st EEG – Non-COVID-19 group (n=46)	p-value	2nd EEG – COVID-19 group (n=49)	2nd EEG – Non-COVID-19 group (n=46)	p-value
Total EEG abnormalities	25 (51.0%)	13 (28.3%)	0.02*	14 (28.6%)	6 (13.0%)	0.05*
Generalized background slowing	13 (26.5%)	5 (10.9%)	0.032*	6 (12.2%)	2 (4.3%)	0.386
Focal slowing	9 (18.4%)	4 (8.7%)	0.045*	5 (10.2%)	2 (4.3%)	0.75
Rhythmic delta patterns	4 (8.2%)	1 (2.2%)	0.476	3 (6.1%)	0 (0.0%)	0.809
Lateralized periodic discharges	6 (12.2%)	2 (4.3%)	0.490	3 (6.1%)	1 (2.2%)	0.540

EEG: Electroencephalogram, COVID-19: Coronavirus Disease 2019

Table 3 Hospital Stay (days) Duration and EEG Findings

EEG finding	COVID-19 group (n=49)	Non-COVID-19 group (n=46)	p-value
Abnormal EEG	20.4 ± 4.4 days	9.6 ± 4.1 days	0.04*
Normal EEG	10.5 ± 4.4 days	6.8 ± 3.2 days	0.048*

EEG: Electroencephalogram, COVID-19: Coronavirus Disease 2019

EEG abnormalities and hospital stay duration

Among COVID-19 patients, those with EEG abnormalities had significantly longer hospital stays (20.4 ± 4.4 days) compared to patients with normal EEG findings (10.5 ± 4.4 days, $p=0.04$). A similar pattern was observed in the non-COVID-19 group, where patients with EEG abnormalities experienced longer hospitalizations (9.6 ± 4.1 days) than those with normal EEGs (6.8 ± 3.2 days, $p=0.048$). These findings suggest a potential association between EEG abnormalities and prolonged clinical recovery, regardless of COVID-19 status. The presence of EEG abnormalities correlated with longer hospital stays, suggesting that neurological involvement in COVID-19 could contribute to prolonged recovery, Table 3.

Subgroup analysis based on demographic and clinical characteristics

A gender-related difference was observed, particularly within the COVID-19 cohort. Female patients with COVID-19 appeared to exhibit higher frequencies of certain EEG abnormalities, including focal slowness (33% compared to 7.1% in males on the initial EEG) and lateralized periodic discharges, Table 4.

In patients with HTN, COVID-19 was associated with a higher observed frequency of EEG abnormalities compared to non-COVID-19 controls, a difference that persisted over time. COVID-19 patients with hypertension demonstrated significantly elevated rates of generalized slowing (58% compared to 20% originally) and rhythmic delta activity, indicative of diffuse encephalopathy. Additionally, abnormalities such as lateralized periodic discharges were observed in 25% of the COVID-19 cohort during the follow-up EEG, which may indicate that the interplay of COVID-19 and HTN could render individuals more susceptible to severe and enduring cerebral

Table 4 Sex subgroup analysis for 1st and 2nd EEG findings in COVID-19 and non-COVID-19

	1st EEG				2nd EEG			
	COVID-19		Non-COVID-19		COVID-19		Non-COVID-19	
	Female N = 21	Male N = 28	Female N = 25	Male N = 21	Female N = 21	Male N = 28	Female N = 25	Male N = 21
Slowing focal	7 (33%)	2 (7.1%)	2 (8.0%)	2 (9.5%)	4 (19%)	1 (3.6%)	1 (4.0%)	1 (4.8%)
Slowing generalized	6 (29%)	7 (25%)	4 (16%)	1 (4.8%)	5 (24%)	1 (3.6%)	2 (8.0%)	0 (0%)
Generalized rhythmic delta activity	3 (14%)	1 (3.6%)	1 (4.0%)	0 (0%)	2 (9.5%)	1 (3.6%)	0 (0%)	0 (0%)
Lateralized rhythmic delta activity	1 (4.8%)	2 (7.1%)	1 (4.0%)	1 (4.8%)	1 (4.8%)	1 (3.6%)	0 (0%)	0 (0%)
Generalized periodic discharges	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)
Lateralized periodic discharges	4 (19%)	2 (7.1%)	1 (4.0%)	1 (4.8%)	2 (9.5%)	1 (3.6%)	0 (0%)	1 (4.8%)
Focal ED	5 (24%)	6 (21%)	3 (12%)	4 (19%)	2 (9.5%)	3 (11%)	2 (8.0%)	1 (4.8%)
Generalized ED	3 (14%)	3 (11%)	0 (0%)	1 (4.8%)	2 (9.5%)	3 (11%)	1 (4.0%)	1 (4.8%)
Multifocal ED	2 (9.5%)	0 (0%)	1 (4.0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	1 (4.8%)
Ictal	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)
Status epilepticus	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)

Table 5 HTN subgroup analysis for 1st and 2nd EEG findings in COVID-19 and non-COVID-19

	1st EEG		2nd EEG	
	COVID-19 N = 12	Non-COVID-19 N = 10	COVID-19 N = 12	Non-COVID-19 N = 10
Slowing focal	3 (25%)	0 (0%)	2 (17%)	0 (0%)
Slowing generalized	7 (58%)	2 (20%)	4 (33%)	0 (0%)
Generalized rhythmic delta activity	2 (17%)	0 (0%)	2 (17%)	0 (0%)
Lateralized rhythmic delta activity	1 (8.3%)	1 (10%)	1 (8.3%)	0 (0%)
Generalized periodic discharges	0 (0%)	0 (0%)	0 (0%)	0 (0%)
Lateralized periodic discharges	3 (25%)	1 (10%)	3 (25%)	1 (10%)
Focal ED	3 (25%)	2 (20%)	2 (17%)	2 (20%)
Generalized ED	2 (17%)	1 (10%)	2 (17%)	1 (10%)
Multifocal ED	2 (17%)	0 (0%)	0 (0%)	1 (10%)
Ictal	0 (0%)	0 (0%)	0 (0%)	0 (0%)
Status epilepticus	0 (0%)	0 (0%)	0 (0%)	0 (0%)

dysfunction, highlighting the potential neurological risk associated with this group. Table 5.

For DM, COVID-19 infection correlated with an increased initial prevalence of EEG abnormalities, notably widespread slowness (40% vs. 20%) and the occurrence of lateralized periodic discharges (20% vs. 0%), indicative of substantial brain dysfunction. Nonetheless, a significant clinical observation is that all these abnormalities completely resolved in the COVID-19 cohort by the second EEG, Table 6.

In patients with pre-existing cardiac conditions, the prevalence and pattern of EEG abnormalities were notably analogous across those with and without COVID-19, a result that contrasts with other high-risk populations. Significantly, abnormalities including focal slowness, lateralized periodic discharges, and epileptiform discharges

Table 6 DM subgroup analysis for 1st and 2nd EEG findings in COVID-19 and non-COVID-19

	1st EEG		2nd EEG	
	COVID-19 N = 5	Non-COVID-19 N = 5	COVID-19 N = 5	Non-COVID-19 N = 5
Slowing focal	0 (0%)	0 (0%)	0 (0%)	0 (0%)
Slowing generalized	2 (40%)	1 (20%)	0 (0%)	0 (0%)
Generalized rhythmic delta activity	0 (0%)	1 (20%)	0 (0%)	0 (0%)
Lateralized rhythmic delta activity	0 (0%)	0 (0%)	0 (0%)	0 (0%)
Generalized periodic discharges	0 (0%)	0 (0%)	0 (0%)	0 (0%)
Lateralized periodic discharges	1 (20%)	0 (0%)	0 (0%)	0 (0%)
Focal ED	1 (20%)	1 (20%)	0 (0%)	0 (0%)
Generalized ED	0 (0%)	0 (0%)	0 (0%)	0 (0%)
Multifocal ED	0 (0%)	0 (0%)	0 (0%)	0 (0%)
Ictal	0 (0%)	0 (0%)	0 (0%)	0 (0%)
Status epilepticus	0 (0%)	0 (0%)	0 (0%)	0 (0%)

remained in the COVID-19 group throughout the second EEG, while they generally resolved in the control group, Table 7.

This subgroup analysis suggests a pattern in individuals with renal disease, indicating that the primary impact of COVID-19 is the persistence rather than the initial occurrence of EEG abnormalities. Although both groups exhibited a modest and comparable proportion of abnormalities in the initial EEG, all abnormalities in the non-COVID-19 group were eliminated by the second recording. The COVID-19 group had new, alarming findings on the follow-up EEG, including lateralized rhythmic delta activity (17%) and sustained lateralized periodic discharges (17%), Table 8.

The examination of patients with pre-existing liver disease indicates that COVID-19 infection correlates with a significantly increased prevalence of encephalopathic

Table 7 Cardiac disease subgroup analysis for 1st and 2nd EEG findings in COVID-19 and non-COVID-19

	1st EEG		2nd EEG	
	COVID-19 N=5	Non-COVID-19 N=12	COVID-19 N=5	Non-COVID-19 N=7
Slowing focal	1 (20%)	2 (17%)	1 (20%)	0 (0%)
Slowing generalized	1 (20%)	3 (25%)	1 (20%)	1 (14%)
Generalized rhythmic delta activity	0 (0%)	0 (0%)	0 (0%)	0 (0%)
Lateralized rhythmic delta activity	0 (0%)	1 (8.3%)	0 (0%)	0 (0%)
Generalized periodic discharges	0 (0%)	0 (0%)	0 (0%)	0 (0%)
Lateralized periodic discharges	1 (20%)	2 (17%)	1 (20%)	1 (14%)
Focal ED	1 (20%)	3 (25%)	1 (20%)	1 (14%)
Generalized ED	1 (20%)	2 (17%)	1 (20%)	1 (14%)
Multifocal ED	1 (20%)	1 (8.3%)	0 (0%)	1 (14%)
Ictal	0 (0%)	0 (0%)	0 (0%)	0 (0%)
Status epilepticus	0 (0%)	0 (0%)	0 (0%)	0 (0%)

Table 8 Renal disease subgroup analysis for 1st and 2nd EEG findings in COVID-19 and non-COVID-19

	1st EEG		2nd EEG	
	COVID-19 N=6	Non-COVID-19 N=8	COVID-19 N=6	Non-COVID-19 N=8
Slowing focal	0 (0%)	1 (13%)	0 (0%)	0 (0%)
Slowing generalized	1 (17%)	0 (0%)	0 (0%)	0 (0%)
Generalized rhythmic delta activity	0 (0%)	1 (13%)	0 (0%)	0 (0%)
Lateralized rhythmic delta activity	0 (0%)	0 (0%)	1 (17%)	0 (0%)
Generalized periodic discharges	0 (0%)	0 (0%)	0 (0%)	0 (0%)
Lateralized periodic discharges	1 (17%)	0 (0%)	1 (17%)	0 (0%)
Focal ED	0 (0%)	1 (13%)	0 (0%)	0 (0%)
Generalized ED	0 (0%)	0 (0%)	1 (17%)	0 (0%)
Multifocal ED	0 (0%)	0 (0%)	0 (0%)	0 (0%)
Ictal	0 (0%)	0 (0%)	0 (0%)	0 (0%)
Status epilepticus	0 (0%)	0 (0%)	0 (0%)	0 (0%)

EEG patterns relative to individuals with other respiratory conditions. COVID-19 patients demonstrated significantly elevated frequencies of both localized and widespread slowing (40% compared to 9.1% and 27% originally) and rhythmic delta activity, which are characteristic indicators of diffuse brain dysfunction, Table 9.

Discussion

EEG findings and COVID-19-related neurological changes

The present study demonstrated that COVID-19 patients exhibited significantly more EEG abnormalities compared to non-COVID-19 patients, both in the first and second EEG assessments. Generalized background

Table 9 Hepatic disease subgroup analysis for 1st and 2nd EEG findings in COVID-19 and non-COVID-19

	1st EEG		2nd EEG	
	COVID-19 N=10	Non-COVID-19 N=11	COVID-19 N=10	Non-COVID-19 N=11
Slowing focal	4 (40%)	1 (9.1%)	2 (20%)	1 (9.1%)
Slowing generalized	4 (40%)	3 (27%)	3 (30%)	1 (9.1%)
Generalized rhythmic delta activity	2 (20%)	0 (0%)	2 (20%)	0 (0%)
Lateralized rhythmic delta activity	2 (20%)	1 (9.1%)	1 (10%)	0 (0%)
Generalized periodic discharges	0 (0%)	0 (0%)	0 (0%)	0 (0%)
Lateralized periodic discharges	2 (20%)	2 (18%)	2 (20%)	1 (9.1%)
Focal ED	2 (20%)	3 (27%)	2 (20%)	3 (27%)
Generalized ED	2 (20%)	1 (9.1%)	2 (20%)	2 (18%)
Multifocal ED	2 (20%)	1 (9.1%)	0 (0%)	1 (9.1%)
Ictal	0 (0%)	0 (0%)	0 (0%)	0 (0%)
Status epilepticus	0 (0%)	0 (0%)	0 (0%)	0 (0%)

slowing and focal slowing were more common in the COVID-19 group ($p=0.032$ and $p=0.045$, respectively). Persistent EEG abnormalities were noted in 28.6% of COVID-19 patients, compared to 13.0% of non-COVID-19 patients ($p=0.05$).

These findings align with previous reports indicating that generalized slowing is a predominant EEG abnormality in COVID-19 patients with encephalopathy. Studies by Verboom et al. [6] and Frontera et al. [7] observed similar EEG patterns, particularly in critically ill COVID-19 patients. However, other studies, such as those by Liotta et al. [8] and Somani et al. [9], suggested that background abnormalities were rare and non-specific to COVID-19 infection.

The presence of rhythmic delta activity and lateralized periodic discharges in COVID-19 patients may reflect the coexistence of systemic inflammation, metabolic disturbances, or other acute illness-related factors that affect cerebral function. Previous literature, including studies by Frontera et al. [7] and Paterson et al. [10], reported rhythmic delta activity in 10–15% of COVID-19 patients with encephalopathy.

Correlation between CO-RADS scale and EEG changes

Interestingly, this study did not find a statistically significant correlation between COVID-19 severity (as assessed by the CO-RADS scale: COVID-19 Reporting and Data System) and EEG abnormalities ($p>0.05$). This contrasts with findings from Hwang et al. [11], who demonstrated a strong correlation between EEG findings and the severity of disease by the CO-RADS scale.

Despite the lack of a significant correlation, the higher frequency of EEG abnormalities in severe COVID-19 cases may indicate that other mechanisms, such as inflammatory cytokine release and microvascular dysfunction, may play a role in neurological involvement [12].

EEG abnormalities and prolonged hospital stay

Patients with abnormal EEG findings had significantly longer hospital stays compared to those with normal EEGs (COVID-19: 20.4 ± 4.4 days vs. 10.5 ± 4.4 days, $p = 0.04$; non-COVID-19: 9.6 ± 4.1 days vs. 6.8 ± 3.2 days, $p = 0.048$). EEG abnormalities in hospitalized patients, particularly those with COVID-19, may reflect acute and subacute cerebral dysfunction, suggested by possible complex interactions among systemic inflammation, metabolic disturbances, microvascular damage, and potentially direct viral effects on neural tissue [13–15]. These pathophysiological mechanisms not only may lead to neurological symptoms but also could extend the duration of clinical recovery, as demonstrated by the observed correlation between EEG pathology and prolonged hospital stays [14]. This phenomenon may continue post-discharge due to persistent or unresolved neuroinflammatory processes and cerebral metabolic disturbances that exceed respiratory improvement, underscoring EEG's significance as a potential indicator of central nervous system (CNS) involvement and the necessity for ongoing neurological monitoring in affected patients [15]. These results are consistent with studies by Atony et al. [16] and Sáez-Landete et al. [17], which found a strong association between EEG abnormalities and prolonged hospitalization in COVID-19 patients.

Clinical advancements

Compared to previous literature, (I) This study distinctly records the continuation of EEG abnormalities, including generalized and focal slowness, rhythmic delta activity, and lateralized periodic discharges, both during acute COVID-19 infection and following clinical recovery before discharge. This observation indicates the persistence of EEG abnormalities beyond the acute illness phase, which may reflect ongoing or slowly resolving cerebral dysfunction. (II) This cohort includes patients with varying respiratory severities, demonstrating that EEG abnormalities can manifest even in moderate COVID-19 instances, in contrast to other investigations that predominantly identified such abnormalities in severe or critically ill patients. This study suggests that EEG abnormalities were not directly associated with the severity of lung disease as assessed by CO-RADS scores. (III) The study reveals a notable linear link between EEG abnormalities and extended hospital stays in patients with both COVID-19 and non-COVID-19

respiratory illnesses. This builds upon previous research that primarily concentrated on EEG data as indicators of acute neurological state instead of being predictors of hospitalization outcomes. (IV) Moreover, comprehensive subgroup analyses demonstrate specific patterns of EEG abnormalities in individuals with comorbidities including hypertension, diabetes, and cardiac, renal, and hepatic disorders, highlighting distinct patterns of EEG abnormalities in these populations, a feature inadequately examined in previous research. (V) The extensive approach utilizing multiple blinded expert reviewers for prolonged EEG monitoring, adhering to approved worldwide norms, enhances the reliability and robustness of findings, establishing a methodological benchmark for future study.

Study limitations

This study faced several limitations that should be considered when interpreting the results. First, the relatively small sample size may limit the generalizability of the findings to broader populations. Second, as a single-center study, there is a potential for selection bias, which may affect the external validity of the results. Third, the follow-up period was restricted to the duration of hospitalization, and long-term outcomes—including the persistence or resolution of EEG abnormalities after discharge—were not assessed. Finally, the observational nature of the study prevents causality or prognostic significance from being inferred. Future multicenter studies with larger cohorts and extended follow-up are warranted to validate and expand upon these findings.

Conclusion

This study highlights the significant association between COVID-19 and EEG abnormalities, particularly generalized background slowing, focal slowing, and periodic discharges. The higher persistence of EEG abnormalities in COVID-19 patients raises the possibility of prolonged cerebral dysfunction, although long-term neurological outcomes were not assessed in this study.

Moreover, the correlation between EEG abnormalities and prolonged hospital stays supports the value of neurological evaluation in COVID-19 patients with altered mental status or cognitive dysfunction. Despite the lack of a direct correlation between CO-RADS severity and EEG findings, the increased prevalence of EEG abnormalities in COVID-19 patients suggests that neurological dysfunction may be influenced by inflammatory, metabolic, or microvascular changes rather than direct pulmonary involvement.

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Author contributions

Mohamed Elsayed Helmy: Study design, data analysis, manuscript drafting. Naglaa Mohamed Ahmed Elkhayat: Supervision, methodology. Ahmed Abd Elmonem Gaber, Yousry Aboelnaga Abdelhamid: Data collection, interpretation. Ahmed Mohamed Hazzou, Mona Mokhtar Wahid El Din: EEG analysis, manuscript revision. All authors have agreed to conditions noted on the Authorship Agreement Form and have read and approved the final version submitted. The content of the manuscript has not been published, or submitted for publication elsewhere.

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Data availability

All raw data will be available on the editor request through communication with the corresponding author.

Declarations

Ethics approval and consent to participate

All procedures performed in the study were in accordance with the ethical standards of the faculty of medicine, Ain Shams university research and ethical committee and with the 1964 Helsinki declaration and its later amendments or comparable ethical standards. Written informed consent was obtained from participants for participation. We obtained approval from research ethics committee no. FWA 000017585. On 10/8/2022 (Approval Number: FMASU MD 180/2022).

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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